

Newspapers and Legislative Effectiveness in the U.S. House

Kisoo Kim*

Craig Volden[†]

Alan Wiseman[‡]

May 4th, 2026

Abstract

Does a watchful local news make legislators focus more on effective lawmaking? We construct a district-congress panel linking local daily newspapers to legislative effectiveness scores (LES) for every U.S. House member from 1873 to 2024. Newspaper presence substantially raises LES, but the effect concentrates among non-Southern Democrats before 1920: a local daily lifts a median legislator to roughly the 68th percentile of effectiveness. For all other groups, the effect is null across 104 years through 2024. Before 1920, Republican access to party machine networks substituted for the newspaper channel, and after 1920 the rise of radio reshaped the information environment. Five mechanism tests (competitiveness, aligned versus opposition press, LES decomposition, seniority persistence, and electoral consequences) favor an information-provision channel over voter accountability: newspapers help legislators learn about their districts rather than discipline them. The value of local news for legislative representation operates on the legislator side, not just the voter side.

Keywords: newspapers, legislative effectiveness, media, Congress, information provision, party machines

*Email: kisoo@virginia.edu

[†]Email: volden@virginia.edu

[‡]Email: alan.wiseman@vanderbilt.edu

1 Introduction

A large literature documents that local media affects political accountability, voter turnout, and policy outcomes (Snyder and Strömberg, 2010; Gentzkow et al., 2011; Gao et al., 2020; Hayes and Lawless, 2018; Peterson, 2021). A parallel literature on congressional behavior, building on Mayhew (1974) and Arnold (1990), examines how legislators respond to electoral incentives and constituent preferences. Yet the intersection of these two literatures—whether and how media coverage makes legislators *more effective*, not just more responsive to electoral pressure, but more productive in the legislative process—remains largely unexplored. Understanding how local media shaped legislative behavior during the formative era of American newspaper expansion is the focus of this article.

We construct a district-congress panel spanning 76 congresses (1873–2024) by merging four data sources: the ICPSR 30261 newspaper panel (Gentzkow et al., 2014a), the Chiou–Goplerud Legislative Effectiveness Scores (Chiou and Goplerud, 2024) for the 43rd–111th Congresses, the Volden–Wiseman LES (Volden and Wiseman, 2014) for the 112th–118th Congresses, and Voteview roll-call data (Lewis et al., 2013). The resulting dataset covers 31,677 legislator-congress observations across 576 unique districts, with time-varying newspaper counts derived from quadrennial directory panels for 1873–1992 and annual operating records from 1994 onward.

Three institutional features of the pre-1920 political environment motivate a search for heterogeneity along era, party, and region. First, newspapers were the dominant—often sole—source of political information before 1920, with no competing broadcast media (Gentzkow et al., 2014a); radio diffused rapidly through the 1920s and reached most American households by 1940 (Strömberg, 2004), with television following mid-century, progressively eroding the newspaper monopoly. Their marginal informational value to legislators and constituents should therefore have been largest before 1920. Second, the pre-1920 party system was organized around machine politics, and Republicans generally

commanded stronger organizational networks than Democrats outside the South (Keller, 1977; McCormick, 1986), leaving non-Southern Democrats more reliant on external information channels. Third, the non-competitive nature of Southern politics under Jim Crow structurally attenuated electoral incentives for legislative effort in that region (Bensel, 2000). These three features jointly suggest that a newspaper effect on legislative effectiveness would bite hardest at their intersection: pre-1920, Democratic, non-Southern.

A two-way fixed effects specification with district and congress fixed effects yields a modest positive association between newspaper presence and LES in the full sample. The eight-way split ($\text{era} \times \text{party} \times \text{region}$) confirms the expected concentration: among non-Southern Democrats serving before 1920, a local daily lifts a median legislator to roughly the 68th percentile of effectiveness. For all other aggregate subgroups—Republicans in both eras, Southern Democrats in both eras, and all post-1920 legislators—the estimated effects are negligible.

The newspaper effect could operate through two channels: voter accountability (newspapers inform voters, who then discipline legislators through electoral pressure) or information provision (newspapers inform legislators themselves about their districts, helping them craft more effective legislation). Five tests generate opposing predictions under each channel. The effect does not concentrate in electorally competitive districts, as accountability would predict. Having any aligned or opposition newspaper raises LES, but adding more aligned papers continues to lift it while adding more opposition papers does not—a pattern inconsistent with opposition scrutiny dominating. Decomposing LES reveals that newspapers raise both bills introduced and their quality. The effect persists across seniority for Democrats but fades after the freshman term for Republicans, consistent with Democrats lacking the party-machine information channel that Republicans access. Electoral channels do not drive the LES effect: controlling for vote margins barely attenuates the coefficient, and newspaper presence does not predict reelection—the LES channel operates independently of electoral accountability.

This paper makes three contributions. First, it provides the first systematic evidence linking historical media environments to legislative productivity, extending the media-and-politics literature beyond electoral outcomes and roll-call voting (Snyder and Strömberg, 2010; Strömberg, 2004; Ferraz and Finan, 2008; Canes-Wrone et al., 2002; Canes-Wrone and Shotts, 2007). Second, it identifies information provision as the operative mechanism, contributing to a growing literature on how information flows shape legislative behavior (Prat and Strömberg, 2013; Cagé, 2020; Miler, 2018). Third, by documenting the role of party machines as information substitutes, it connects the media literature to the political history of American party organizations (Aldrich, 1995; Cox and McCubbins, 2005; McCormick, 1986), offering an explanation for why the newspaper–effectiveness relationship disappears after 1920 and does not extend to Republicans.

2 Historical Background

2.1 The Newspaper Era, 1870–1920

The half-century between the Civil War and World War I constituted the golden age of American newspaper expansion. Between 1870 and 1910, the number of daily newspapers in the United States more than quadrupled, driven by urbanization, rising literacy rates, technological innovations in printing, and the expansion of advertising revenue (Gentzkow et al., 2014a; Schudson, 1978; Baldasty, 1992). County-level coverage shows comparable growth: only 8% of US counties contained a daily newspaper in 1869, rising to 24% by 1892 and 34% by 1920. In the data used here, 73% of congressional districts contained at least one daily newspaper during the Gilded Age (1873–1892), rising to 90% by the Progressive Era (1893–1922); by 1910, the average district was served by over seven daily newspapers.

Critically, newspapers during this period were the *sole* mass medium. Commercial radio broadcasting did not begin until the 1920s, and television arrived decades later.

For a legislator seeking to communicate with constituents, or for voters seeking information about their representative’s activities, the local newspaper was the only available channel. This media monopoly gives the pre-1920 period unique analytical leverage: any relationship between newspaper presence and legislative behavior cannot be attributed to substitution toward or away from competing media.

2.2 Partisan Press and Information Flows

Most newspapers before 1920 were openly partisan, frequently serving as organs of local party organizations (McCormick, 1986; Gerring, 1998; Kaplan, 2002; Summers, 1994). A district’s media environment was thus characterized not merely by the number of papers, but by their partisan composition. Democratic newspapers covered Democratic legislators favorably and transmitted information about constituent needs through a sympathetic lens; Republican papers did the same for Republican incumbents. The transition from this partisan model toward the “objective” professional journalism of the twentieth century—a transition that Kaplan (2002) traces to the 1865–1920 period—coincides with the post-1920 disappearance of the newspaper effect documented in Section 5.

This partisan structure generates a specific prediction about the information provision mechanism. If newspapers help legislators learn about constituent preferences and local conditions, the effect should flow primarily through *aligned* (same-party) newspapers, which have both the incentive and the editorial orientation to transmit constituency-relevant information. Under an accountability mechanism, by contrast, *opposition* newspapers—which scrutinize and criticize the incumbent—should be the primary driver. The analysis in Section 6 exploits this distinction.

This framework also implies a functional parallel between the pre-1920 opposition newspaper and the post-1920 professional newspaper: both operate as non-aligned observers rather than same-party information conduits. If the LES effect flows primarily through aligned partisan channels, we should observe muted effects from opposition

newspapers within the pre-1920 period, and muted effects from newspapers overall after 1920. Section 6 and Section 8 jointly test this prediction, treating the post-1920 null not as a separate phenomenon but as the natural extension of the aligned/opposition asymmetry documented before 1920.

2.3 Party Machines as Information Networks

The late 19th century was also the era of party machines—hierarchical organizations that mobilized voters, distributed patronage, and coordinated legislative agendas (McCormick, 1986; Cox and McCubbins, 2005). Crucially, the two major parties differed in organizational capacity. The Republican Party, anchored by the business community and industrial interests, maintained more professionalized and centralized organizational networks in the North. The Democratic Party outside the South was a looser coalition of urban ethnic organizations and rural populists, with weaker centralized information infrastructure. Within the South, by contrast, the Democratic Party exercised an organizational monopoly so complete that competitive two-party dynamics were structurally absent (Key, 1949; Mickey, 2015; Bensel, 2000)—a condition with direct consequences for how newspapers could and could not function as information channels.

Direct quantitative measurement of party organizational capacity across the pre-1920 period is not available, but historical scholarship documents a clear organizational asymmetry. Republican state organizations in the industrial North—most prominently Thomas Platt’s machine in New York and Matthew Quay’s (later Boies Penrose’s) machine in Pennsylvania—built strong party organizations tied to business interests that sustained coordinated patronage and convention networks across decades (Keller, 1977). Republican dominance of this period rested on a regionally coherent industrial coalition (Bensel, 2000) that Democratic organizations outside the South did not match. Within the South, the Democratic Party’s organizational monopoly operated through entirely different mechanisms unrelated to competitive two-party information flows (Key, 1949;

Mickey, 2015).

This organizational asymmetry has a direct implication for the newspaper–effectiveness relationship. If party machines served as alternative conduits of district information—conveying constituent demands, policy priorities, and local conditions to legislators—then legislators embedded in strong party organizations had less need for newspaper-mediated information. Republican legislators, supported by more robust organizational networks, already possessed much of the intelligence that newspapers could provide. Democratic legislators outside the South, lacking comparable organizational support, were more dependent on newspaper coverage for the information necessary to craft effective legislation. The null Republican effect documented in this paper is thus not an anomaly but a theoretically predicted consequence of differential party organization.

2.4 The 1920 Structural Break

Several developments around 1920 fundamentally altered the relationship between media and legislative politics. The 19th Amendment (1920) expanded and diversified the electorate. The rise of radio in the 1920s broke the newspaper monopoly on mass communication. The professionalization of Congress—including the expansion of committee staff, the Legislative Reference Service (1914), and standardized office resources—provided legislators with institutional sources of information that partially substituted for newspaper coverage. And the progressive reform movement weakened party machines, reducing the organizational asymmetry that conditioned the partisan differential.

These concurrent changes make it difficult to attribute the post-1920 disappearance of the newspaper effect to any single factor. In Section 8, we examine three leading explanations—women’s suffrage, congressional professionalization, and media environment changes—and find that the evidence is most consistent with the emergence of alternative information sources rendering the newspaper channel redundant.

3 Data

3.1 Data Sources

The analysis combines five principal data sources.

Newspaper Data. The district-level newspaper panel combines two sources. For the 43rd–102nd Congresses (1873–1992), county-level daily newspaper counts and circulation come from ICPSR Study 30261, “United States Newspaper Panel, 1869–2004” (Gentzkow et al., 2014b), compiled from Ayer’s and Rowell’s directories at presidential-year (quadrennial) intervals. We aggregate city-level observations to counties, map counties to congressional districts using the population-weighted crosswalks of Ferrara et al. (2021), and apply linear interpolation between directory years to obtain congress-level measures. For the 103rd–118th Congresses (1993–2024), we use an annual district-level panel of 1,970 U.S. dailies constructed by integrating the Gentzkow–Shapiro base panel with newspaper closure records from Gao et al. (2020), nondaily-transition data from the UNC News Desert project (Abernathy, 2020), and hand-collected web verification for 1992–2024, mapping newspapers to congressional districts via the same crosswalks. The primary treatment variable is `has_newspaper`, an indicator equal to one if the district has at least one daily newspaper in a given congress.

Legislative Effectiveness Scores. We measure legislative productivity using two complementary scores that target the same underlying concept but compute it differently. For the 112th–118th Congresses (2011–2024), we use the Volden–Wiseman Classic LES (Volden and Wiseman, 2014; Volden et al., 2013), which weights each sponsored bill by the legislative stage it reaches (introduced, action in committee, beyond committee, chamber passage, enacted) interacted with its substantive significance (commemorative, substantive, substantive-and-significant) as coded by the Congressional Research Service (CRS), then normalizes to a congress mean of one. For the 43rd–111th Congresses (1873–2010),

we use the Chiou–Goplerud extended effectiveness score (Chiou and Goplerud, 2024), which applies an item response theory model to twelve historically comparable raters—committee reports and hearings, roll-call votes, notable amendments, bill versions, floor-speech mentions, chamber passage, minor-bill keywords, and others—to estimate a continuous latent “notability” for each of 1.1 million introduced bills, then aggregates to the member-congress level as the log of the sum of exponentiated bill notabilities. Despite these substantial differences in construction, the two measures correlate at 0.89 on the log scale in the 1973–2010 overlap (Chiou and Goplerud, 2024). Because the two measures operate on different scales (CG mean ≈ 3.30 , VW mean ≈ 0.99) and have modestly different within-congress variances, we standardize LES within each Congress to mean zero and standard deviation one before pooling. This within-Congress z -score is the dependent variable throughout the analysis.

Voteview. Roll-call voting data, including DW-NOMINATE ideology scores, party codes, and biographical information, come from the Voteview database (Lewis et al., 2013; Poole and Rosenthal, 1997). These data provide party affiliation, seniority, committee assignments, and ideological positions for matching with the newspaper and LES panels.

Election Returns. Election returns come from two complementary sources. District-level returns from Tamas (2021) (1840–1974) provide two-party vote shares for contested House races and support the main vote share analysis (Section 6) and electoral competitiveness measures. County-level returns from Clubb et al. (2006) (ICPSR 8611, 1840–1972), aggregated to congressional districts via the embedded district-number crosswalk, provide both the pre-1920 House vote margins and the presidential two-party Democratic vote share used in the electoral competitiveness tests (Section 6).

3.2 Panel Construction

The unit of observation is a legislator-congress pair. We merge the data sources by congress, state, and district number, with newspapers crosswalked to each congress's district boundaries, yielding a panel of 31,677 observations spanning the 43rd through 118th Congresses (1873–2024) across 576 unique state-district combinations. The original panel (1873–2010, 28,574 observations) uses CG LES and ICPSR 30261 newspaper data. The extension (2011–2024, 3,103 observations) uses VW LES and annual newspaper tracking data mapped via county-to-district crosswalks.

Table 1 reports summary statistics for the full sample and the key analytic subsample. In the full panel, 89% of legislator-congresses have at least one local daily newspaper. The median district has approximately 3 daily newspapers (mean: 4.3), with substantial variation across eras: mean newspaper counts rise from 3.1 during the Gilded Age to 5.9 during the Progressive Era, then gradually decline as consolidation reduces the number of independent dailies, reaching a mean of 2.4 by the 118th Congress (2023–2024).

The pre-1920 non-Southern Democratic subsample—the focus of the main results—comprises 2,323 legislator-congress observations across 263 unique districts. In this subsample, 76% of observations have at least one newspaper, and the mean LES is 3.54 (SD = 1.28) on the raw scale; all regression analyses use the within-Congress standardized measure.

3.3 Newspaper Variation

Figure 1 displays the evolution of newspaper presence over the study period. Panel A plots the mean number of dailies per district (intensive margin); Panel B tracks the share of districts with at least one daily (extensive margin); Panel C plots first-entry and last-exit events per congress. Several features are noteworthy. First, there is substantial cross-sectional variation in every congress, with some districts having no daily newspaper and

Table 1: Descriptive Statistics

	Full Sample		NS-Dem Pre-1920	
	Mean	SD	Mean	SD
<i>Panel A: Newspaper variables</i>				
Has newspaper	0.88	0.33	0.76	0.43
N dailies	4.51	7.82	3.57	4.91
<i>Panel B: Legislative effectiveness</i>				
LES	3.46	1.31	3.54	1.28
N bills introduced	42.1	51.3	43.7	36.2
Vote participation	0.984	0.031	0.984	0.029
<i>Panel C: Legislator characteristics</i>				
Democrat	0.53	0.50	1.00	0.00
Seniority (terms)	3.24	3.01	1.95	1.43
Freshman	0.29	0.45	0.39	0.49
Majority party	0.54	0.50	0.58	0.49
Committee chair	0.06	0.23	0.05	0.22
DW-NOMINATE	0.40	0.16	0.29	0.12
Observations	31,654		2,323	
Congresses	43–118		43–66	
Districts	576		263	

Notes: Full sample covers the 43rd–118th Congresses (1873–2024). LES for 1873–2010 is the Chiou–Goplerud extended score; for 2011–2024 the Volden–Wiseman Classic LES is used. LES values are reported here on their raw scales; all regression analyses use a within-Congress standardized measure (mean 0, SD 1 per Congress). NS-Dem Pre-1920 restricts to non-Southern Democrats in the 43rd–66th Congresses. Newspaper data from ICPSR 30261 (1873–1992, quadrennial, interpolated) and an integrated annual panel (1994–2024).

others having more than 20. Second, the most dramatic expansion occurs in the pre-1920 period, when newspapers are entering districts for the first time—the primary source of identifying variation in this era, as Panel C makes visible. Post-1920 identification instead draws on exits and gradual count changes, an asymmetry consistent with the structural transformation of newspapers discussed in Section 2. Third, the post-1920 period is characterized by gradual consolidation, with few new entries but steady exit as newspapers close or merge. By the 118th Congress (2023–2024), the mean number of dailies per district has fallen to 2.4, with 8.7% of districts having no daily newspaper—more than double the share during the 1950s–1970s.

The identifying variation comes primarily from two sources: (1) districts that gain their first daily newspaper (163 first-entry events over the panel), and (2) districts that experience changes in the number of dailies. The first-entry events are concentrated in the pre-1920 period (101 of 163 occur before the 53rd Congress, 1892), providing the sharpest identification for the stock measure.

4 Empirical Strategy

The causal claim rests on *within-district* variation in newspaper presence over time, holding constant all time-invariant district characteristics and all nationwide trends. We triangulate across five estimators—TWFE, first-difference, Gardner (2022) two-stage, Sun–Abraham (2021), and matched difference-in-differences—and address spatial dependence with Conley (1999) standard errors in robustness. The key identifying assumption is that, conditional on district and congress fixed effects and observable legislator controls, the timing of newspaper entry into a district is not systematically correlated with unobserved determinants of legislative effectiveness.

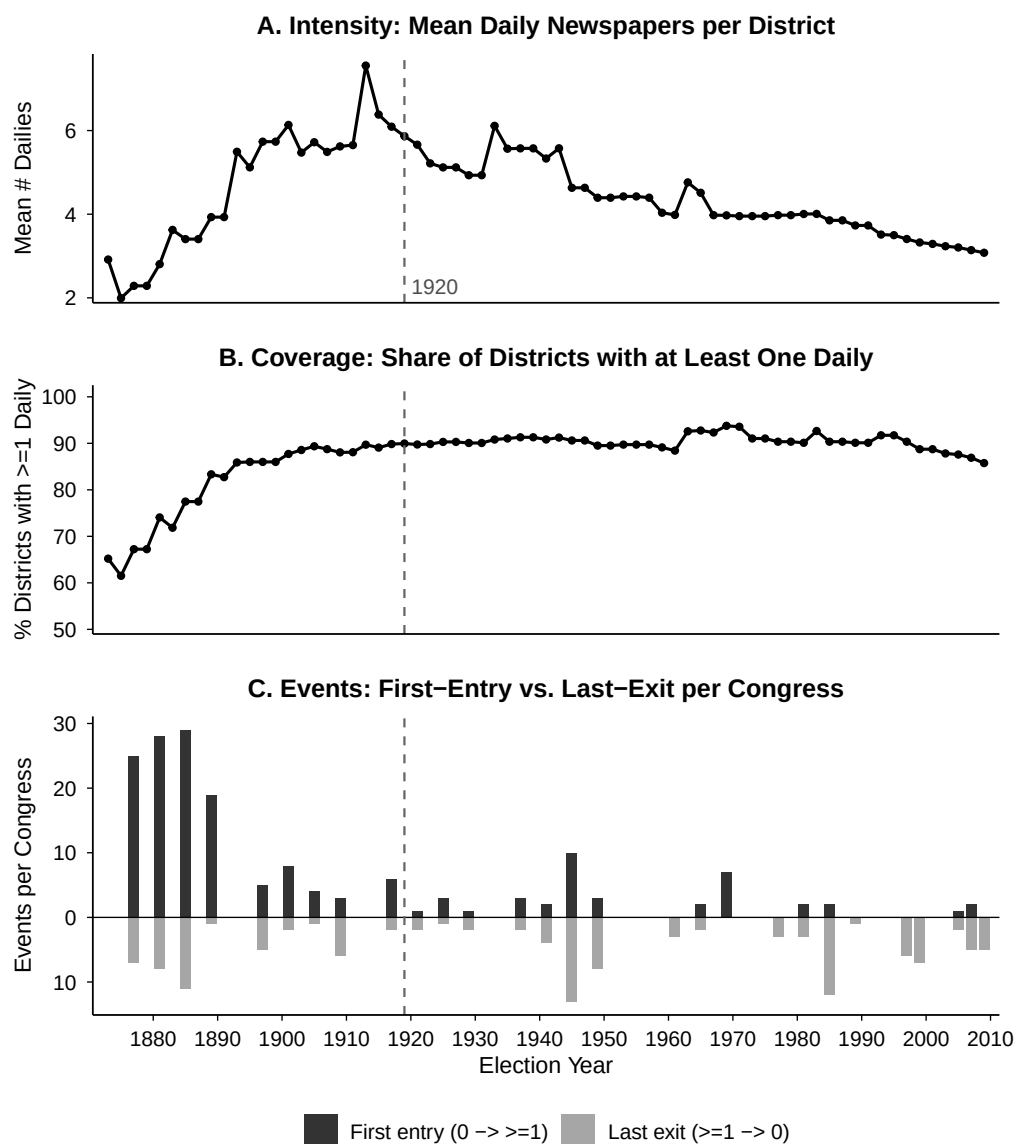


Figure 1: Newspaper Presence Across Congressional Districts, 1873–2010

Notes: Three-panel display of district-level newspaper variation. Panel A: mean daily newspapers per district (intensive margin). Panel B: share of districts with at least one daily (extensive margin). Panel C: first-entry events (districts gaining their first daily, dark bars above zero) and last-exit events (districts losing their last daily, light bars below zero) per congress. The vertical dashed line marks 1920 (the 67th Congress), which separates the pre-1920 analysis period from the post-1920 placebo period. Data from ICPSR 30261, interpolated to congress level.

4.1 Baseline Specification

We estimate the effect of newspaper presence on legislative effectiveness using a two-way fixed effects (TWFE) model:

$$\text{LES}_{ict} = \alpha_i + \gamma_t + \beta \cdot \text{Newspaper}_{ict} + \mathbf{X}'_{ict} \boldsymbol{\delta} + \varepsilon_{ict} \quad (1)$$

where LES_{ict} is the imputed legislative effectiveness score for legislator i in district (absorbed by α_i) during congress t ; α_i and γ_t are district and congress fixed effects, respectively; Newspaper_{ict} is a measure of local newspaper presence—either a binary indicator for any daily newspaper or the count of dailies depending on specification; and $\mathbf{X}_{ict}$ includes time-varying controls: seniority, a freshman indicator, majority party status, and committee chair status. Standard errors are clustered at the district level throughout.

The district fixed effects absorb all time-invariant district characteristics (geography, baseline economic development, political culture), while congress fixed effects absorb nationwide trends in LES and newspaper counts. The coefficient β is thus identified from *within-district* variation in newspaper presence over time, conditional on national trends.

The Chiou–Goplerud (CG, C43–C111) and Volden–Wiseman Classic (VW, C112–C118) LES measures operate on different scales (CG within-Congress mean ≈ 3.23 , SD ≈ 1.27 ; VW mean ≈ 0.99 , SD ≈ 1.15)¹. To address this, the dependent variable is within-Congress standardized: $\text{LES}_{ict}^{\text{std}} = (\text{LES}_{ict} - \bar{\text{LES}}_t) / \sigma_t$, where $\bar{\text{LES}}_t$ and σ_t are the Congress- t mean and standard deviation across all House members. This removes cross-Congress scale differences while preserving within-Congress variation; coefficients are interpretable as standard deviation units of within-Congress LES. Congress fixed effects are retained throughout: within-Congress z-scoring removes mean shifts in the outcome but does not absorb Congress-level confounds of the treatment.

¹The VW sample mean falls slightly below the normalized value of 1.0 because Alaska and Hawaii are excluded from Congresses 112–116 due to the county-to-congressional-district crosswalk lacking coverage for these states in that period. These 16 member-congress observations (0.5% of the VW sample) are excluded from all analyses using the extended panel.

4.2 Treatment Measures

The primary treatment is the binary stock measure `has_newspaper`, which captures the extensive margin: whether a district has *any* daily newspaper. We also examine intensive-margin measures: $\log(1 + n_{\text{dailies}})$, the raw count of dailies, and event-based measures (first entry, any increase, any decrease, last exit). The stock measure is preferred because (1) it captures the most policy-relevant margin (having some local coverage versus none), (2) it is less sensitive to interpolation error, and (3) the first-entry events provide the cleanest identifying variation.

4.3 Subgroup Analysis

The full-sample TWFE pools across eras, parties, and regions that theory predicts will exhibit different effects. We therefore estimate Equation 1 separately for each cell of an eight-way partition: $\{\text{pre-1920, post-1920}\} \times \{\text{Democrat, Republican}\} \times \{\text{South, non-South}\}$. This partition is motivated by three theoretical priors: (1) the pre-1920 newspaper monopoly on information, (2) the partisan asymmetry in party organization, and (3) the non-competitive nature of Southern elections. The resulting eight coefficients are presented as a coefficient plot (Figure 2) that constitutes the central result of the paper.

4.4 Identification Concerns

The main threat to identification is time-varying confounders that correlate with both newspaper entry and legislative effectiveness. Urbanization is the most salient concern: growing districts may simultaneously attract newspapers and send more effective legislators to Congress. We address this in three ways. First, we control directly for district population and urban share using decennial census data (Section 7). Second, we implement propensity score matching on pre-treatment characteristics. Third, the first-difference estimator, which uses only within-district changes between consecutive congresses, pro-

vides an alternative identification strategy that is robust to slower-moving confounders.

The staggered adoption of newspapers also raises concerns about negative weighting in TWFE with heterogeneous treatment effects (Goodman-Bacon, 2021; Sun and Abraham, 2021). We implement several diagnostics: the Gardner two-stage estimator (Borusyak et al., 2024), a switcher-only TWFE restricted to districts that change treatment status, and the Sun and Abraham interaction-weighted estimator (Sun and Abraham, 2021). These are reported in Section 7.

5 Results

5.1 Full Sample Estimates

Table 2 reports the full-sample TWFE results. The binary stock measure yields $\hat{\beta} = 0.096$ ($p = 0.029$) with controls, a statistically significant positive association. Without controls, the estimate is slightly larger at 0.111 ($p = 0.019$). The intensive-margin measures are consistent: $\log(1 + n_{\text{dailies}})$ is significant (0.040, $p = 0.036$), and the raw count is highly significant (0.004, $p < 0.001$). These estimates use the extended panel (1873–2024, $N = 31,648$); the original panel (1873–2010) yielded similar but less precise estimates ($\hat{\beta} = 0.083$, $p = 0.082$ with controls).

These full-sample estimates are informative but pool across subpopulations with fundamentally different political environments. The next two subsections decompose this average.

5.2 Theory-Motivated Heterogeneity

The theoretical framework in Section 2 predicts that the newspaper effect should be strongest where three conditions hold simultaneously: (1) newspapers are the only information source (pre-1920), (2) the legislator’s party lacks strong organizational networks

Table 2: Newspaper Presence and Legislative Effectiveness: Full Sample TWFE

	(1)	(2)	(3)	(4)	(5)	(6)
Has newspaper	0.106* (0.047)	0.093* (0.043)				
log(1+NP)			0.041* (0.019)	0.041* (0.019)		
NP count					0.004*** (0.001)	0.004*** (0.001)
Seniority		−0.005 (0.006)		−0.005 (0.006)		−0.005 (0.006)
Freshman		−0.379*** (0.017)		−0.379*** (0.017)		−0.379*** (0.017)
Majority party		0.333*** (0.018)		0.333*** (0.018)		0.333*** (0.018)
Committee chair		0.816*** (0.039)		0.816*** (0.039)		0.816*** (0.039)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Congress FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	31,648	31,648	31,648	31,648	31,648	31,648

Notes: Dependent variable is within-Congress standardized LES (mean 0, SD 1 per Congress; CG measure for 1873–2010, VW Classic for 2011–2024). All specifications include district and congress fixed effects. Standard errors clustered at the district level in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

(Democrats outside the South), and (3) elections are contested (non-South). Figure 2 reports the eight-way split, and the results are striking.

Consistent with our theoretical prediction, pre-1920 non-Southern Democrats exhibit a large, precisely estimated effect ($\hat{\beta} = 0.335$, $SE = 0.094$), representing 0.34 standard deviations of within-Congress LES, or roughly the difference between a legislator in the 50th and 68th percentiles of effectiveness. All seven remaining cells—Republicans in both eras and regions, Southern Democrats in both eras, and post-1920 non-Southern Democrats—yield point estimates close to zero and statistically insignificant.

The three-way interaction ($\text{has_newspaper} \times \text{democrat} \times \text{pre-1920}$) is statistically significant in the pooled model, confirming that the conditional pattern is not an artifact of

Table 3: Heterogeneity: Era $\times$ Party $\times$ Region

Era	Party	Region	$\hat{\beta}$	SE	N
Pre-1920	Democrat	Non-South	0.335***	0.094	2,323
Pre-1920	Democrat	South	0.155	0.122	1,993
Pre-1920	Republican	Non-South	0.066	0.098	4,157
Pre-1920	Republican	South	-0.180	0.425	221
Post-1920	Democrat	Non-South	0.064	0.082	8,272
Post-1920	Democrat	South	0.224	0.123	4,185
Post-1920	Republican	Non-South	0.067	0.083	8,522
Post-1920	Republican	South	0.028	0.215	1,876

Notes: Each row reports a separate TWFE regression of within-Congress standardized LES on `has_newspaper` within the indicated subgroup using the extended panel (1873–2024). Pre-1920 results are identical to the original panel. All specifications include district and congress fixed effects and controls (seniority, freshman, majority party, committee chair). Standard errors clustered at the district level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

subsample comparisons. The non-Southern restriction is supported by a negative interaction between newspaper presence and a South indicator among pre-1920 Democrats ($p = 0.061$).

5.3 Dose-Response and Treatment Intensity

Within the pre-1920 non-Southern Democratic subsample, the dose-response pattern further supports a causal interpretation. Compared to districts with no newspapers, the LES premium rises from 0.164 (not significant) for districts with a single daily, to 0.359 ($p < 0.001$) for two to three dailies, to 0.378 ($p = 0.001$) for four to six, and to 0.472 ($p < 0.001$) for seven or more. This pattern—a sharp initial jump at the threshold of two newspapers, followed by a gradual plateau—is consistent with a model where the first few newspapers provide the bulk of the information, with diminishing returns thereafter. For Republicans, no dose category shows a significant effect.

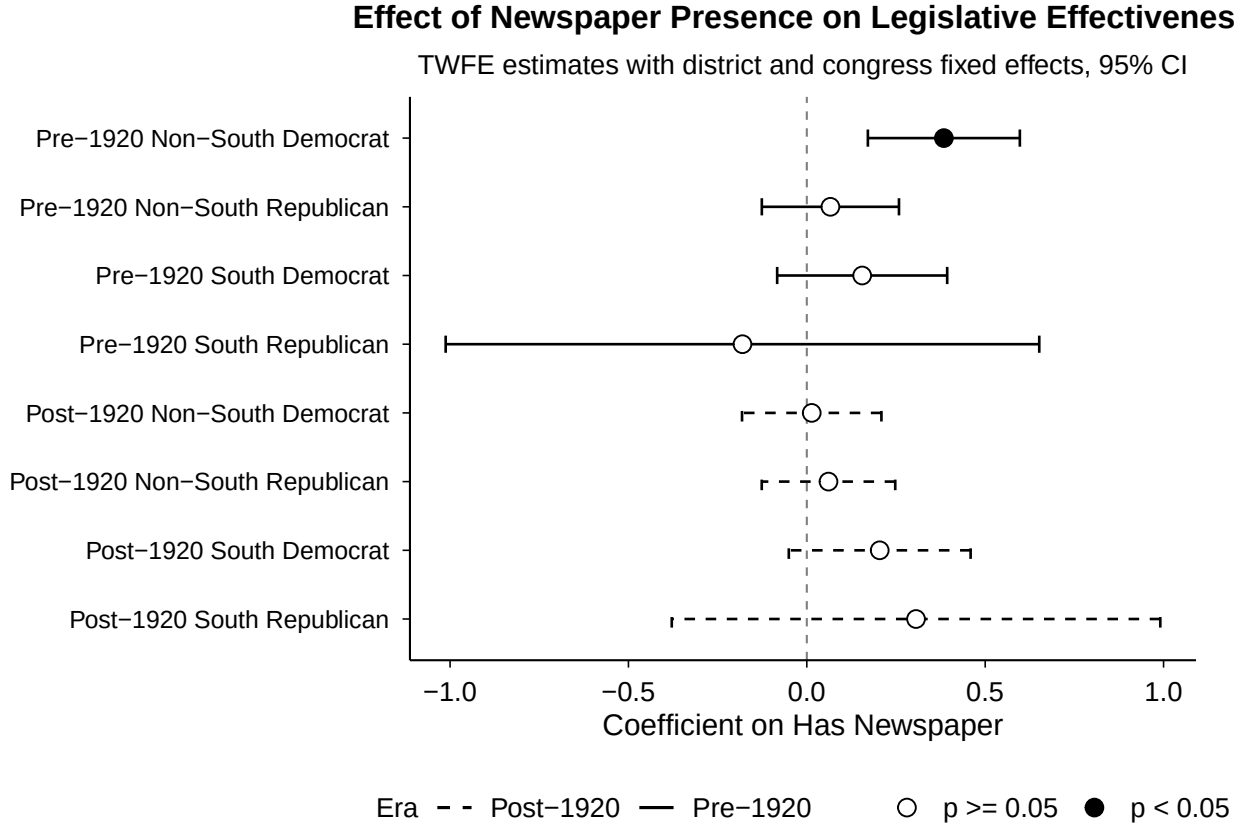


Figure 2: Newspaper Effect on LES: Eight-Way Heterogeneity

Notes: Each point represents a separate TWFE regression of within-Congress standardized LES on `has_newspaper` within the indicated subgroup (era $\times$ party $\times$ region). Horizontal bars are 95% confidence intervals based on district-clustered standard errors. All specifications include district and congress fixed effects and legislator-level controls.

6 Mechanism: Information Provision vs. Accountability

The positive effect of newspapers on Democratic LES is consistent with two broad mechanisms. Under *voter accountability*, newspapers enable voters to monitor their representative, generating electoral pressure that incentivizes legislative effort (Snyder and Strömberg, 2010; Besley and Burgess, 2002; Ferraz and Finan, 2008). Under *information provision*, newspapers transmit constituency-relevant information to legislators—voter preferences, local needs, policy priorities—enabling them to craft more effective legislation even absent electoral pressure (Auslen, 2025). These mechanisms generate divergent predictions across five testable dimensions.

6.1 Test 1: Electoral Competitiveness

Accountability predicts that the newspaper effect should be stronger in competitive districts, where electoral consequences of shirking are most severe. Information provision predicts that the effect should operate regardless of competitiveness, since information helps legislative production independent of reelection incentives.

We assess electoral competitiveness using two direct measures: the Democratic two-party presidential vote share (aggregated from ICPSR 08611 county returns to congressional districts) and the lagged House two-party vote margin (Tamas, 2021). Districts above the sample median on each measure are classified as “safe.” Using presidential vote share, the newspaper effect in competitive districts is $\hat{\beta}_{\text{competitive}} = 0.155$ ($p = 0.245$) and in safe districts $\hat{\beta}_{\text{safe}} = 0.033$, with interaction $p = 0.494$. Using prior House vote margin: $\hat{\beta}_{\text{competitive}} = 0.067$ ($p = 0.576$) and $\hat{\beta}_{\text{safe}} = 0.273$, interaction $p = 0.113$. Neither proxy produces a statistically significant interaction. This pattern favors information provision over accountability.

6.2 Test 2: Aligned vs. Opposition Press

Accountability predicts that opposition newspapers—which scrutinize the incumbent—should drive the effect. Information provision predicts that aligned newspapers—which transmit sympathetic constituency information—should matter more. Among pre-1920 non-Southern Democrats, extensive-margin effects are roughly symmetric across the two paper types: aligned (Democratic) papers yield $\hat{\beta} = 0.258$ ($p < 0.001$) and opposition (Republican) papers yield $\hat{\beta} = 0.251$ ($p = 0.003$) under district and congress fixed effects (Figure 3; Appendix Table A.8). At the intensive margin, the number of party-aligned dailies has a positive (marginally significant) effect on LES ($\hat{\beta} = 0.050$, $p = 0.058$), while the number of opposition dailies has a *negative* (marginally significant) effect ($\hat{\beta} = -0.030$, $p = 0.075$). The entry of a new aligned newspaper is associated with higher LES (+0.121,

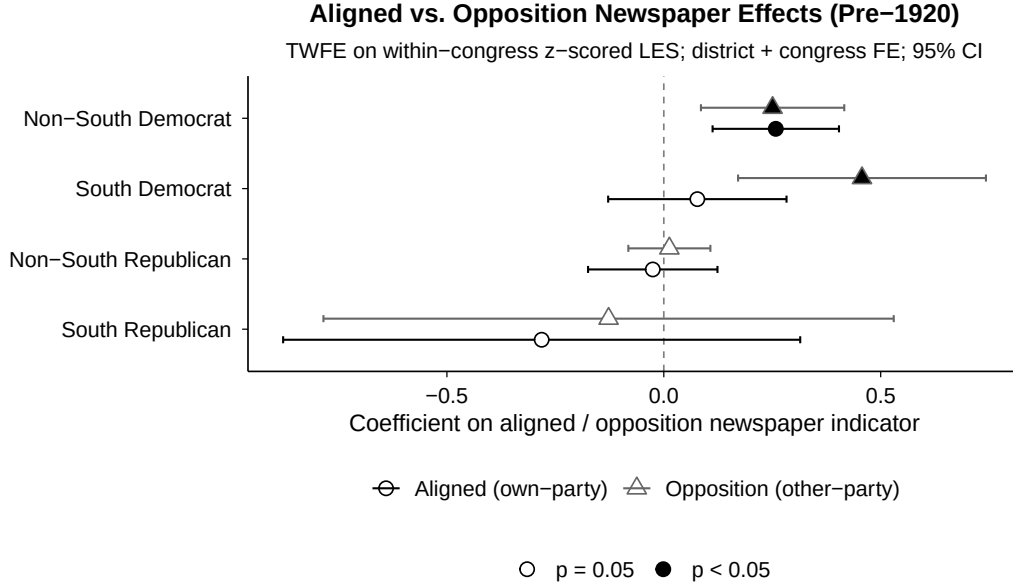


Figure 3: Aligned vs. Opposition Newspaper Effects, Pre-1920

Notes: Each point represents a separate TWFE regression of within-Congress standardized LES on the indicated newspaper indicator within the labeled subgroup. “Aligned” is the own-party paper indicator (*has_dem_paper* for Democrats, *has_rep_paper* for Republicans); “opposition” is the other-party indicator. Horizontal bars are 95% confidence intervals based on district-clustered standard errors. All specifications include district and congress fixed effects and the legislator-level controls used in the main text. The Non-South Democrat row recovers the roughly symmetric aligned/opposition pattern reported in the text. The South Democrat row shows a markedly different pattern in which only the opposition (Republican) paper is significant ($\hat{\beta} = 0.457, p = 0.002$), while the aligned (Democratic) paper is null ($\hat{\beta} = 0.077, p = 0.46$); this asymmetry is consistent with opposition newspapers serving as the sole accountability channel in the non-competitive Jim Crow South. Republicans are null in both columns and both regions.

$p = 0.239$), while the entry of an opposition newspaper is associated with *lower* LES ($-0.049, p = 0.616$), though neither is individually significant. These patterns are more consistent with information provision—both paper types transmit constituency-relevant information, with a slight edge for aligned papers that have the editorial incentive to cover the legislator’s constituency—than with pure accountability, which would predict a dominant role for opposition scrutiny.

6.3 Test 3: LES Decomposition

We decompose the newspaper effect into its legislative components. If newspapers work through accountability (pure effort incentive), the effect should appear primarily on the extensive margin—more bills introduced—with less impact on legislative quality. If newspapers work through information provision, the effect should extend to *quality* as well, since better information allows legislators to craft bills that advance further through the process.

For pre-1920 non-Southern Democrats, newspapers increase both the number of bills introduced (+8.4 bills, $p = 0.018$) and roll-call voting participation (+0.9 percentage points, $p < 0.001$). A mediation analysis reveals that 31% of the total LES effect is mediated by quantity (more bills), while 69% operates through the direct (quality) channel. The residual LES effect after controlling for $\log(1 + n_{\text{bills}})$ remains significant ($\hat{\beta} = 0.099$, $p = 0.037$). This evidence is more consistent with information provision, which predicts effects on both margins.

Table 5 summarizes the five mechanism tests and their implications.

6.4 Test 4: Seniority Persistence and Party Machines

If the partisan asymmetry reflects differential access to party machine information, we should observe a specific seniority pattern. For Republicans, whose party machine provides an alternative information source, the newspaper effect should appear only among freshmen (who have not yet been integrated into the organizational network) and fade with seniority. For Democrats, who lack comparable organizational support, the effect should persist across seniority levels.

Figure 4 presents the results. Among Republicans, there is a marginally significant freshman effect (+0.203, $p = 0.057$) that disappears—and even reverses—at higher seniority levels. Among Democrats, the effect is positive and significant for freshmen (+0.284,

Table 4: LES Decomposition: Quantity vs. Quality (NS-Dem Pre-1920)

Outcome	$\hat{\beta}$	SE	p
<i>Panel A: NS-Democrat</i>			
LES (composite)	0.338***	0.096	0.001
N bills (quantity)	8.368*	3.515	0.018
log(1 + N bills)	0.250**	0.083	0.003
LES per bill (quality)	0.007	0.013	0.575
Vote participation	0.009***	0.003	0.001
<i>Panel B: Mediation</i>			
Total effect	0.338***	0.096	0.001
Indirect (via N bills)	0.106		
Direct (quality)	0.231**	0.078	0.003
% mediated by quantity		31.4%	
<i>Panel C: Republican (comparison)</i>			
LES (composite)	0.028	0.086	0.748
N bills (quantity)	2.779	3.096	0.370
Vote participation	0.002	0.004	0.598

Notes: All specifications include district and congress fixed effects and controls. Standard errors clustered at the district level. Pre-1920 non-South subsamples. The indirect effect is computed as the product of the newspaper→N bills coefficient and the N bills→LES coefficient conditional on newspaper presence. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table 5: Mechanism Scorecard: Information Provision vs. Accountability

Test	Account.	Info Prov.	Evidence
Competitive > safe seats	Predicted	Not predicted	Safe $\approx$ competitive (pres. $p_{\text{int}} = 0.49$; House $p_{\text{int}} = 0.11$)
Opposition press drives effect	Predicted	Not predicted	Aligned press +, opposition –
Quality channel (LES n_bills)	Not predicted	Predicted	69% direct/quality channel
Rep effect persists w/ seniority	No prediction	Not predicted	Rep fades after freshman
Vote margin mediates LES	Predicted	Not predicted	Attenuation = -1.1%

All five tests favor information provision.

Notes: “Account.” = voter accountability mechanism; “Info Prov.” = information provision mechanism. Each row summarizes the prediction of each mechanism and the empirical evidence from Section 6. See text for full details.

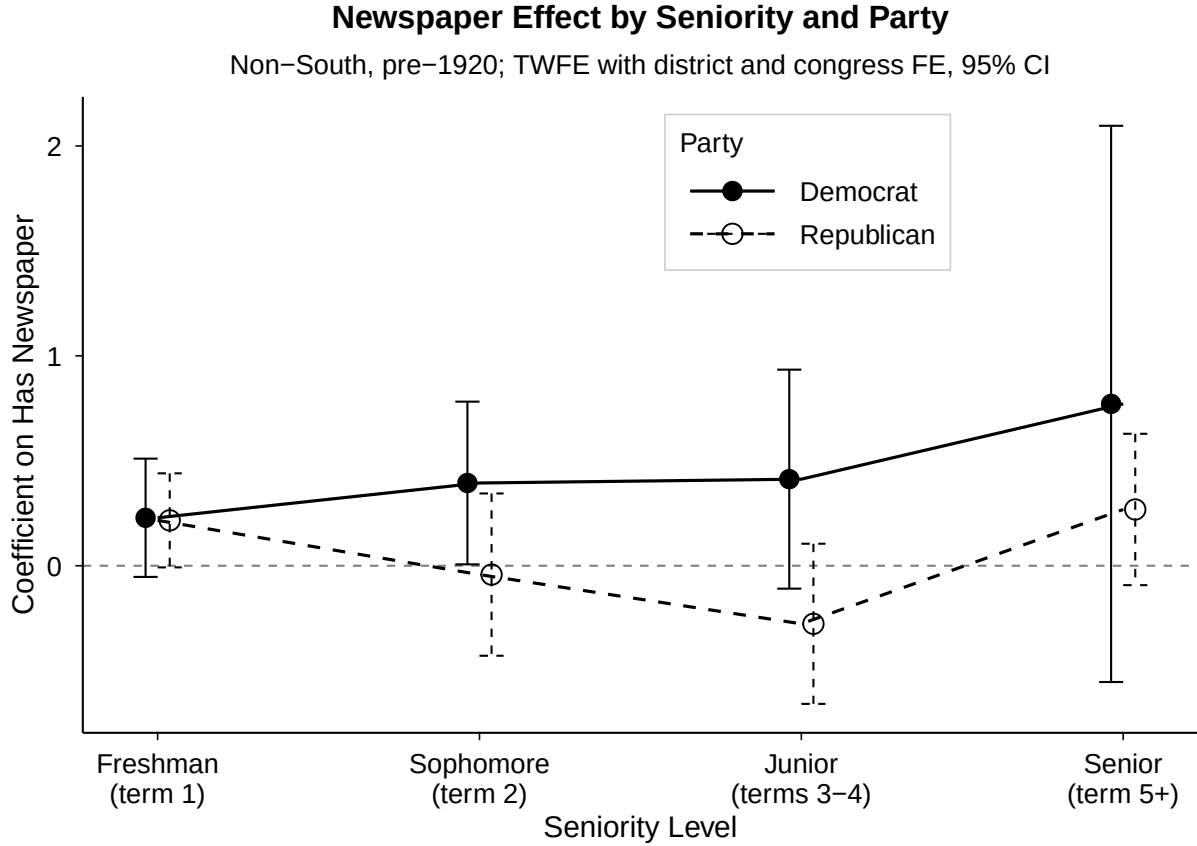


Figure 4: Newspaper Effect by Seniority and Party (NS Pre-1920)

Notes: The figure shows the estimated effect of `has_newspaper` on LES at each seniority level, separately for Democrats (filled circles, solid line) and Republicans (open circles, dashed line) in the non-South pre-1920 subsample. Vertical bars are 95% confidence intervals. The Democratic effect persists across seniority levels, while the Republican effect fades after the freshman term.

$p = 0.035$), sophomores ($+0.390$, $p = 0.030$), and juniors ($+0.461$, $p = 0.054$). The formal interaction test confirms the divergence: the `has_newspaper` $\times$ `seniority` interaction is negative for Republicans (-0.053 , $p = 0.083$) and positive (though not significant) for Democrats ($+0.042$, $p = 0.203$).

This pattern is uniquely consistent with the party machine hypothesis. If the Republican null were due to lower baseline LES sensitivity, we would not expect a significant freshman effect. Instead, the freshman-only effect for Republicans suggests that newspapers *do* provide useful information, but this information is quickly superseded by party organizational channels as the legislator gains seniority.

6.5 Test 5: Electoral Consequences and the Accountability Null

The foundational case for newspaper effects on legislative behavior, developed in Snyder and Strömberg (2010), Besley and Burgess (2002), and Ferraz and Finan (2008), runs through voter accountability: newspapers enable voters to monitor incumbents, punish poor performers electorally, and thereby induce effort *ex ante*. If this channel operates here, we should observe three implications among pre-1920 non-Southern Democrats: (i) newspapers should make elections more competitive (raising the marginal cost of shirking); (ii) more effective legislators should be more likely to win reelection; and (iii) the LES effect should be substantially mediated by electoral competitiveness.

The evidence on the first implication is consistent with accountability: newspapers *reduce* Democratic two-party vote share by 3.7 percentage points ($p < 0.001$) and shrink absolute margins by 3.4 percentage points ($p < 0.001$). Races become measurably more competitive when newspapers arrive.

The evidence on the second implication, however, is starkly inconsistent with accountability. Regressing reelection on LES (controlling for district and congress fixed effects, seniority, freshman status, majority status, and committee chair status) yields a coefficient of +0.002 ($p = 0.962$) among pre-1920 non-Southern Democrats—a precise zero. Voters did not reward legislative effectiveness with higher reelection probabilities. Indeed, no reelection regression in any specification—full sample, pre-1920 only, by party, or by region—produces a significant LES-reelection relationship. The electoral sanction that accountability theory requires is absent.

The evidence on the third implication also contradicts accountability. Adding lagged absolute margin as a control to the baseline LES regression attenuates the newspaper coefficient by only 1.1% (from 0.492 to 0.487), and the margin coefficient itself is small and insignificant ($p = 0.660$). The electoral competitiveness that newspapers induce does not mediate the LES effect.

Taken together, the electoral evidence presents a pattern that is difficult to reconcile

with any version of the accountability mechanism. Newspapers reshape electoral competition without translating legislative effort into electoral reward, and the LES effect operates independently of the electoral channel entirely. This is not merely an absence of supporting evidence for accountability; it is positive evidence against it, generated precisely in the historical setting—the pre-radio, pre-television, partisan-press era—where the accountability channel has been thought to operate most strongly (Snyder and Strömberg, 2010). The finding directs attention to the alternative information-provision channel, in which newspapers transmit constituency-relevant information to legislators themselves and thereby improve legislative output independently of whether voters notice. A complementary argument in modern state legislatures is developed by Auslen (2025), who finds that press coverage strengthens roll-call responsiveness to district opinion most in less-professionalized legislatures where staff-based information substitutes are scarce—a heterogeneity pattern consistent with newspapers functioning as an information subsidy for legislators, though Auslen does not directly test the accountability channel the way the reelection null reported here does.

6.6 Mechanism Summary

Table 5 presents a scorecard across the five tests. In every case, the evidence favors information provision over accountability. Newspapers help Democratic legislators by providing constituency-relevant information that improves both the quantity and quality of their legislative output. The partisan asymmetry is explained by Republicans’ access to party machine information networks that substitute for the newspaper channel.

7 Robustness

7.1 Alternative DID Estimators

Recent methodological advances have highlighted potential biases in TWFE estimators when treatment effects are heterogeneous across cohorts or time (Goodman-Bacon, 2021; Sun and Abraham, 2021; Callaway and Sant’Anna, 2021). To address these concerns, we implement four alternative estimators. Table 6 reports the results.

The Gardner two-stage DID estimator (Borusyak et al., 2024), which estimates unit and time fixed effects using only untreated observations, yields $\hat{\beta} = 0.494$ ($p < 0.001$, $N = 1,191$)—considerably larger than the baseline TWFE, suggesting that if anything, TWFE understates the true effect. A switcher-only TWFE restricted to districts that change newspaper status produces $\hat{\beta} = 0.398$ ($p < 0.001$, $N = 1,127$). The first-difference estimator, which uses only within-district changes between consecutive congresses, gives $\hat{\beta} = 0.219$ ($p = 0.039$, $N = 2,075$).

The Gardner event study (Figure 5) shows no significant pre-treatment coefficients, supporting the parallel trends assumption. Post-treatment coefficients become significant at periods 3–4 after newspaper entry, consistent with a delayed effect as newly available information is gradually incorporated into the legislative process.

7.2 Interpreting the Sun–Abraham Null

The Sun–Abraham (2021) interaction-weighted estimator produces a null estimate ($\hat{\beta} = 0.076$, $p = 0.710$) when `first_entry` is used as the treatment timing variable. Because S&A has become a standard diagnostic for staggered-adoption designs, this null warrants direct engagement. We argue here that the S&A result is an artifact of the empirical setting—specifically, the concentration of treated cohorts in the Gilded Age and the scarcity of never-treated districts—rather than evidence against a newspaper effect.

Table 6: Alternative DID Estimators (NS-Dem Pre-1920)

Estimator	$\hat{\beta}$	SE	p	N
TWFE (baseline)	0.335***	0.094	< 0.001	2,323
TWFE (switchers only)	0.398***	0.097	< 0.001	1,127
First difference	0.219*	0.106	0.039	2,075
Gardner did2s	0.494***	0.138	< 0.001	1,191
Sun & Abraham (first entry)	0.076	0.204	0.710	2,323
Sun & Abraham (any increase)	0.311*	0.137	0.024	2,323
<i>Placebo: Post-1920 NS-Dem</i>				
TWFE	0.026	0.101	0.795	7,110

Notes: All estimators target the effect of newspaper presence on within-Congress standardized LES for the NS-Dem pre-1920 subsample, except the post-1920 placebo. The Gardner did2s estimator uses only untreated observations to estimate fixed effects. “Switchers only” restricts to districts that change newspaper status during the sample. Standard errors clustered at the district level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

The cohort concentration problem. The S&A estimator proceeds in three steps: (i) estimate cohort-specific average treatment effects (CATTs) for each treatment-timing cohort using that cohort’s treated observations and never-treated observations as controls; (ii) aggregate the CATTs into a pooled effect using weights proportional to cohort size. When treated units are concentrated in one or two early cohorts, steps (i) and (ii) yield imprecise estimates for two reasons. First, the individual CATTs rely on small within-cohort samples. Second, the aggregation weights concentrate in those imprecisely estimated cohorts, so the pooled estimate inherits their noise. Among pre-1920 non-Southern Democratic districts, 60 of 67 `first_entry` events occur during the Gilded Age (1873–1892). The S&A estimator is therefore asked to identify an average effect using a single dominant cohort, a setting in which it is known to lose power (Roth et al., 2023; de Chaisemartin and D’Haultfœuille, 2020).

The never-treated scarcity problem. S&A identifies CATTs using never-treated observations as the comparison group. For pre-1920 non-Southern Democrats, only 7 of 263 districts remain never-treated throughout the sample. This comparison group is too small

Event Study: Newspaper Entry and Legislative Effectiveness

Gardner (2022) two-stage DID, Non-South Democrats pre-1920

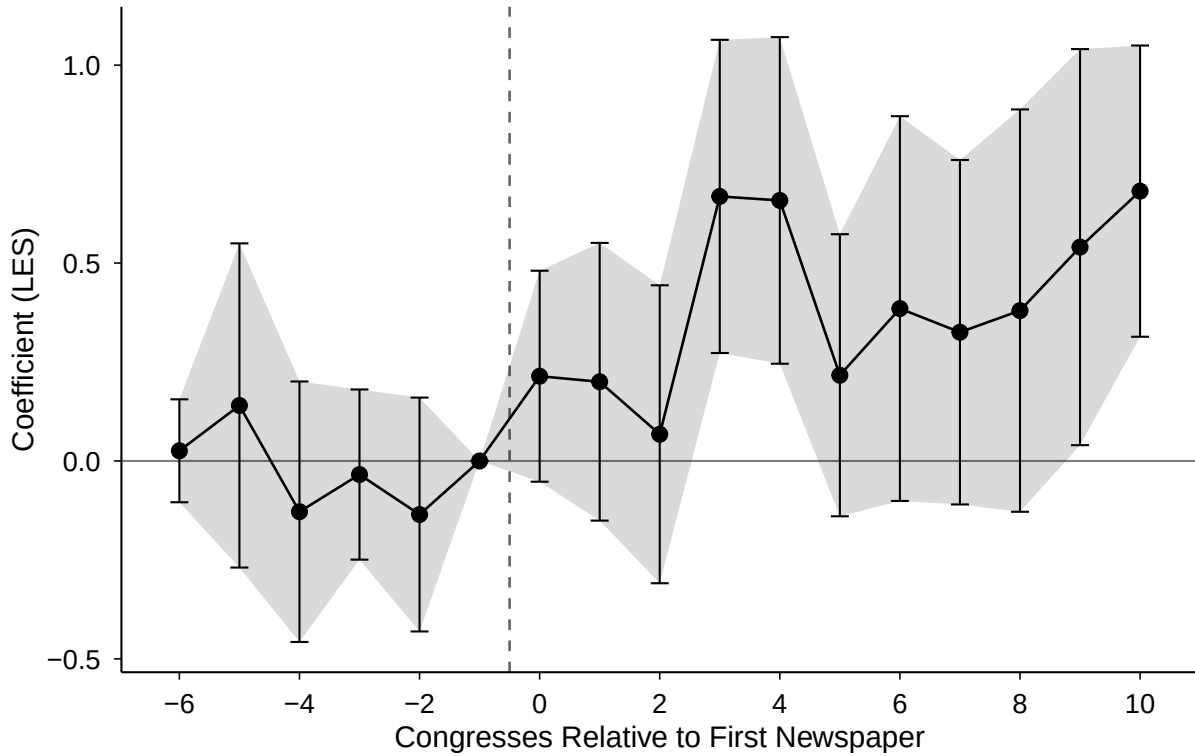


Figure 5: Gardner Two-Stage Event Study (NS-Dem Pre-1920)

Notes: The figure plots dynamic treatment effect estimates from the Gardner (did2s) two-stage estimator for non-Southern Democrats in the pre-1920 period. The reference period is $t - 1$ (one congress before newspaper entry). Vertical bars are 95% confidence intervals. Pre-treatment coefficients are individually insignificant, supporting the parallel trends assumption.

to deliver precise CATT estimates. By contrast, the Gardner (2022) two-stage estimator uses all not-yet-treated and never-treated observations (including post-treatment periods of other cohorts under the assumption of homogeneous treatment effects) to estimate unit and time fixed effects in the first stage, yielding substantially more precise estimates in this setting.

Evidence that power, not absence of effect, drives the null. Three observations support this interpretation. First, re-running S&A with `any_increase` as the treatment indicator—a specification that classifies districts as treated when they gain any addi-

tional newspaper, enlarging the pool of treated units and reducing cohort concentration—recovers a positive and significant estimate ($\hat{\beta} = +0.311$, $p = 0.024$). Second, the Gardner event study (Figure 5) shows flat pre-trends and significant positive post-treatment effects at $t + 3$ and later, a pattern consistent with a real lagged effect that S&A’s cohort-weighted aggregation cannot efficiently capture. Third, a minimum detectable effect calculation using the S&A standard error ($SE = 0.204$) implies an MDE of approximately 0.57 LES points at 80% power—larger than the TWFE, first-difference, and did2s point estimates, confirming that S&A is underpowered to detect effects of the observed magnitude in this setting. Moreover, the dose-response pattern in Section 5 shows that the LES premium jumps sharply at the two-to-three newspaper threshold rather than at a district’s first newspaper, which suggests that `first_entry` may capture the wrong margin for this setting: the effect operates primarily at the intensive margin, which S&A pooled over first-entry cohorts cannot isolate.

Recommended interpretation. The choice of a staggered-adoption estimator is not mechanical; it involves trade-offs between bias under treatment-effect heterogeneity (which S&A addresses well) and variance under cohort concentration (where S&A loses). Roth et al. (2023) emphasize that the appropriate estimator depends on the data-generating process. In settings like this one—dominant cohort, small never-treated group—they recommend alternatives such as Gardner (2022) and Borusyak et al. (2024), both of which we implement and both of which produce positive, significant estimates. We therefore interpret the body of evidence as supporting a positive newspaper effect, with the S&A null reflecting a known weakness of that estimator under this specific cohort structure rather than a substantive contradiction of the other estimates.

Dynamic structure. For transparency, Appendix Table A.10 reports the period-by-period did2s and Sun–Abraham coefficients for $\tau \in [-5, +5]$ around first newspaper entry, along with a binned-persistence regression that partitions post-treatment observa-

tions into short ($\tau \in [0, 2]$), medium ($\tau \in [3, 5]$), and long ($\tau \geq 6$) windows. Individual period coefficients are insignificant, and pairwise tests cannot reject equality across the three post-treatment bins ($p = 0.76$ for short vs. medium, $p = 0.25$ for short vs. long, $p = 0.29$ for medium vs. long). The data are consistent with a flat positive level shift on the order of the TWFE stock estimate but underpowered to adjudicate the precise dynamic profile. This is exactly the pattern predicted by the cohort-concentration argument above.

7.3 Urbanization Confound

Newspaper presence is correlated with urbanization, raising the concern that the estimated effect reflects district development rather than media per se. Using decennial census data (ICPSR 2896), we construct district-level population and urban share measures for the pre-1920 period and add them as controls. The newspaper coefficient is attenuated but remains strongly significant: from a baseline of 0.356 to 0.289 when all three urbanization controls (log population, urban share, and log farm value) are included simultaneously, retaining 81% of the effect ($p = 0.003$). Notably, adding urban share alone raises the coefficient slightly (to 0.397), reflecting that newspaper coverage is negatively correlated with urban share ($r = -0.26$) in this sample—newspapers were concentrated in small and mid-sized towns rather than large industrial cities. The overall pattern confirms that time-varying urbanization does not account for the estimated newspaper effect.

7.4 Propensity Score Matching

We implement two matching strategies to address concerns about non-comparability of treated and control observations. Cross-sectional propensity score matching at each congress—matching newspaper and non-newspaper districts on seniority, ideology, and urban share—yields $\hat{\beta} = 0.340$ ($p = 0.011$, $N = 858$). District-level matching on pre-

treatment characteristics for switcher districts yields a larger estimate of $\hat{\beta} = 0.468$ ($p = 0.001$, $N = 669$). Both approaches confirm that the baseline estimate is not driven by compositional differences between newspaper and non-newspaper districts.

7.5 Placebo Tests

Two placebo tests support a causal interpretation. First, a post-1920 placebo repeats the main specification for non-Southern Democrats in the post-1920 period (1920–2024), where theory predicts a null effect. Over 104 years of post-1920 data, the estimate is near zero ($\hat{\beta} = 0.064$, $p = 0.441$, $N = 8,272$), consistent with the theoretical prediction and ruling out persistent unobserved heterogeneity. The broader post-1920 placebo for all legislators is also null ($\hat{\beta} = 0.063$, $p = 0.159$, $N = 22,895$).

Second, lead newspaper tests examine whether *future* newspaper presence predicts *current* LES. In a horse-race regression including both current and one-period-ahead newspaper indicators, only the current measure is significant ($\hat{\beta}_{\text{current}} = 0.290$, $p = 0.005$; $\hat{\beta}_{\text{lead}} = 0.082$, $p = 0.452$). The insignificance of the lead term supports the interpretation that newspapers cause higher LES, rather than both being driven by underlying trends.

8 The Post-1920 Disappearance

The restriction of the newspaper effect to the pre-1920 period raises the question of why the relationship dissolved. We evaluate three candidate explanations.

Women’s Suffrage. The 19th Amendment (1920) dramatically expanded the electorate, potentially disrupting the information channel by introducing a large cohort of new voters whose preferences were less well-served by existing newspapers. However, analysis exploiting state-level variation in the timing of women’s suffrage shows no systematic relationship between suffrage adoption and the newspaper–LES effect. The pre/post-1920

Table 7: Robustness Checks (NS-Dem Pre-1920)

Specification	$\hat{\beta}$	SE	p	N
Baseline	0.335***	0.094	< 0.001	2,323
+ All urbanization controls	0.289***	0.095	0.003	2,032
+ Lagged vote margin	0.363***	0.104	< 0.001	2,046
Cross-sectional PSM	0.340**	0.132	0.011	858
District-level matched DID	0.468***	0.133	0.001	669
Excluding KY & MO	0.379***	0.106	< 0.001	1,822
<i>Placebo tests</i>				
Post-1920 NS-Dem	0.026	0.101	0.795	7,110
Lead newspaper (horse race)	0.082	0.109	0.452	—

Notes: All specifications target the effect of `has_newspaper` on LES for non-Southern Democrats in the pre-1920 period. Baseline includes district and congress FE with controls. Cross-sectional PSM matches treated and control observations on seniority, ideology, and urban share within each congress. District-level matched DID matches switcher districts to control districts on pre-treatment characteristics. Standard errors clustered at the district level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

break is not explained by suffrage timing.

Congressional Professionalization. The creation of the Legislative Reference Service (1914), expansion of committee staff, and standardization of office resources provided legislators with institutional information sources. If professionalization substituted for newspaper-mediated information, we would expect the newspaper effect to attenuate gradually as these institutions developed. However, the break is relatively sharp rather than gradual, and the magnitude of the professionalization changes in the early 20th century appears insufficient to fully account for the disappearance.

Media Environment Changes. The arrival of commercial radio in the 1920s broke the newspaper monopoly on mass communication. Simultaneously, the wire service model homogenized newspaper content, reducing the distinctive local information that had made newspapers valuable for legislative production. The combination of competing

media and content homogenization is most consistent with the timing and sharpness of the post-1920 break.

Direct Constituent Information. Beyond institutional substitutes within Congress and competing mass media, the post-1920 period saw the gradual emergence of channels through which legislators could learn about constituent preferences without the newspaper as intermediary. Scientific opinion polling, pioneered commercially in the 1930s and routinized in congressional offices by the postwar period, offered a direct alternative to inferring district sentiment from local press coverage. The expansion of district offices and standardized constituent service operations during the 1960s and 1970s further reduced reliance on newspapers as the primary conduit of district information. None of these channels existed at scale before 1920. Their cumulative effect was to make the local newspaper one of many information sources rather than the principal one—weakening the marginal informational value that gave pre-1920 newspapers their distinctive role in legislative production.

These explanations are not mutually exclusive, and the available evidence does not permit a definitive adjudication. The most likely interpretation is that multiple developments around 1920—media competition, content standardization, congressional professionalization, the emergence of direct constituent-information channels (polling, district offices), and the weakening of party machines—collectively rendered the newspaper channel redundant.

9 Discussion

9.1 Interpreting the Event Study

The Gardner event study in Figure 5 shows flat pre-treatment coefficients consistent with parallel trends and positive post-treatment point estimates that are individually impre-

cise (see Appendix Table A.10 for the full period-by-period coefficients from both *did2s* and Sun–Abraham). The imprecision at individual horizons does not undermine the aggregate finding. As Roth et al. (2023) emphasize, event studies are designed to test for pre-trends, not to estimate dynamic treatment effects precisely, and are typically underpowered for the latter purpose. The positive average effect is confirmed by the TWFE, first-difference, and Gardner aggregate estimates, all of which are statistically significant. The detailed power argument is developed in Section 7.2.

9.2 Geographic Spillovers

A concern with district-level analysis is that newspapers may exert spillover effects across adjacent districts, either biasing the own-district estimate upward (knowledge diffusion from neighbors) or downward (competition for reader attention). We address this concern directly using distance-based spatial controls. For each district-congress, we compute the mean newspaper exposure of all other districts within radii of 100, 250, and 500 kilometers—using Haversine great-circle distance between district centroids constructed from the Ferrara–Testa–Zhou county-to-district crosswalk—and include this neighbor measure alongside the own indicator in the regression. Appendix Table A.9 reports six specifications.

Three patterns emerge. First, the own-newspaper coefficient is essentially unchanged across all radii (+0.348 at 100 km, +0.345 at 250 km, +0.351 at 500 km, all $p < 0.01$), indicating that the main finding is not an artifact of spatial aggregation. Second, the neighbor coefficient is near zero at the tightest geographic scale (100 km: +0.137, $p = 0.47$) and becomes statistically null once state-by-congress fixed effects are included (250 km neighbor with state $\times$ congress FE: +0.53, $p = 0.40$), consistent with the interpretation that apparent spillovers at wider radii capture state-level confounders rather than genuine cross-district information flows. Third, a placebo using *lead* neighbor exposure (one-congress-ahead) is insignificant (-0.78 , $p = 0.15$), while contemporaneous neighbor exposure is marginally

positive—a pattern inconsistent with a pre-trend in LES that tracks future newspaper growth.

Replacing district-clustered standard errors with Conley (1999) spatial heteroskedasticity- and autocorrelation-consistent errors (Conley, 1999) confirms this conclusion ($p = 0.0001$; see Appendix Table A.9).

9.3 The Republican Industrial Anomaly

One subgroup of pre-1920 Republicans shows a significant positive newspaper effect: legislators representing districts with above-median industrial employment, concentrated in the Big Four states (New York, Pennsylvania, Ohio, and Illinois) that dominated early twentieth-century American manufacturing. The effect in this subgroup is $\hat{\beta} = +0.41$ ($p = 0.032$), roughly comparable in magnitude to the pre-1920 Democratic effect.

Two considerations qualify the tension this creates with the party-machine hypothesis. First, the affected subgroup is narrow: it requires the conjunction of Republican, pre-1920, and above-median industrial employment, and it does not generalize to other Republican subgroups. Second, the party-machine substitution argument is about the *relative* information value of newspapers versus organizational channels, not an absolute substitutability. Industrial districts presented legislators with an unusually heterogeneous set of constituent interests—manufacturing labor, immigrant communities, emerging service sectors, railroad interests—whose specific policy demands may have exceeded the capacity of centralized party organizations to aggregate. In this reading, the party machine provided adequate constituency intelligence for the modal Republican district but not for the most complex ones, leaving a residual information role for newspapers. Rather than falsifying the substitution story, this subgroup locates its boundary: party machines substituted for newspapers where constituency demands were coherent enough for the organization to process, and failed to substitute where demands were too heterogeneous.

This account is tentative, and richer data on industrial composition and party organi-

zational capacity would be required to test it directly. The broader partisan asymmetry—Republican nulls in most cells, Democratic positive effects across the non-South—is unaffected.

9.4 Contemporary News Deserts: A Null Extension

Does the modern collapse of local newspapers produce an effect in reverse—newspaper loss reducing legislative effectiveness today as newspaper gain raised it before 1920? We test this directly using the full range of the Volden–Wiseman Classic LES measure (C93–C118, 1973–2024), which spans the onset and acceleration of local news decline. Merging VW LES with the newspaper panel and re-estimating the baseline TWFE yields $\hat{\beta} = +0.012$ (SE = 0.051, $p = 0.815$, $N = 11,462$). Splitting at C112 (2011), the approximate onset of digital disruption, yields $\hat{\beta} = +0.012$ ($p = 0.841$) for 1973–2010 and $\hat{\beta} = +0.052$ ($p = 0.747$) for 2011–2024. The news desert hypothesis receives no support across any sub-period.

This null is consistent with the paper’s theoretical framework rather than anomalous to it. The pre-1920 effect required newspapers to serve as the *primary* information channel between district and legislator. Modern legislators operate in an information-rich environment—congressional staff, wire services, cable television, and digital media all provide constituency-relevant intelligence even where local newspapers have closed. The marginal information value of a local daily is far smaller when it competes with many substitutes than when it was the sole conduit. Binary newspaper presence, our measure, may also be too coarse for the modern question: in the contemporary era, the relevant margin is likely coverage *intensity* rather than mere presence, consistent with Auslen (2025)’s finding that state legislators are more responsive when newspapers are specifically incentivized to cover their districts. The pre-1920 result and the modern null are, in this reading, two sides of the same theoretical claim.

10 Conclusion

This paper provides the first systematic evidence that local newspaper presence affected legislative effectiveness in the U.S. House of Representatives, using a district-congress panel spanning 76 congresses (1873–2024). The effect is concentrated among non-Southern Democrats serving before 1920, for whom gaining a local daily newspaper increased legislative effectiveness by 0.34 standard deviations. The null effect for all post-1920 subgroups persists across 104 years of placebo data (1920–2024), providing strong evidence against persistent unobserved heterogeneity. Five mechanism tests consistently identify information provision—rather than voter accountability—as the operative channel: newspapers helped Democratic legislators learn about constituent needs and craft more effective legislation, rather than disciplining them through electoral pressure.

The partisan asymmetry illuminates the role of party organizations as information intermediaries. Republicans, embedded in stronger party machine networks, already had access to the constituency intelligence that newspapers provided. Democrats, who lacked comparable organizational support outside the South, were more dependent on the newspaper channel. The disappearance of the effect after 1920 is consistent with the emergence of competing media, the professionalization of Congress, and the erosion of party machines collectively rendering the newspaper information channel redundant.

These findings reframe rather than extend contemporary debates about local news decline. The closure of local newspapers in the 21st century (Darr et al., 2018, 2021; Gao et al., 2020; Hayes and Lawless, 2018, 2021; Peterson, 2021; Mathews, 2022) is typically studied through its voter-side effects on information and electoral accountability. Our historical evidence identifies a complementary legislator-side channel, but one that was operative precisely because newspapers were the *primary* constituency-information conduit before 1920. Modern legislators operate in a fundamentally different environment: congressional staff, wire services, cable news, and digital media all provide constituency-

relevant intelligence even where local papers have closed. Our own null across the full Volden–Wiseman period (C93–C118) reflects this substitution logic rather than revealing an absence of harm. The contemporary concern is therefore about coverage *intensity*, not presence: if the relevant modern margin is how much locally-focused reporting legislators receive—rather than whether a daily newspaper exists—then shrinking newsroom capacity may quietly erode the channel even where a masthead survives. Contemporary evidence supports this framing: Auslen (2025) finds that state legislators are more responsive to district opinion when newspapers are specifically incentivized to cover them, with effects concentrated precisely where information substitutes are weakest—the same structural configuration our historical evidence identifies as generating legislative effects.

References

- Abernathy, P. M. (2020). News deserts and ghost newspapers: Will local news survive? Technical report, University of North Carolina at Chapel Hill, Hussman School of Journalism and Media.
- Aldrich, J. H. (1995). *Why Parties? The Origin and Transformation of Political Parties in America*. University of Chicago Press, Chicago.
- Arnold, R. D. (1990). *The Logic of Congressional Action*. Yale University Press, New Haven.
- Auslen, M. (2025). Local news and policy responsiveness in the states. Working paper, University of Texas at Austin.
- Baldasty, G. J. (1992). *The Commercialization of News in the Nineteenth Century*. University of Wisconsin Press, Madison.
- Bensel, R. F. (2000). *The Political Economy of American Industrialization, 1877–1900*. Cambridge University Press, New York.
- Besley, T. and Burgess, R. (2002). The political economy of government responsiveness: Theory and evidence from India. *Quarterly Journal of Economics*, 117(4):1415–1451.
- Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting event-study designs: Robust and efficient estimation. *Review of Economics and Statistics*, 106(2):516–534.
- Cagé, J. (2020). *The Price of Democracy: How Money Shapes Politics and What to Do about It*. Harvard University Press, Cambridge, MA.
- Callaway, B. and Sant’Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

- Canes-Wrone, B., Brady, D. W., and Cogan, J. F. (2002). Out of step, out of office: Electoral accountability and house members' voting. *American Political Science Review*, 96(1):127–140.
- Canes-Wrone, B. and Shotts, K. W. (2007). When do elections encourage ideological rigidity? *American Political Science Review*, 101(2):273–288.
- Chiou, F.-Y. and Goplerud, M. (2024). Effective lawmaking across congressional eras. *The Journal of Politics*, 86(3):997–1012.
- Clubb, J. M., Flanigan, W. H., and Zingale, N. H. (2006). Electoral data for counties in the United States: Presidential and congressional races, 1840–1972. Inter-university Consortium for Political and Social Research [distributor], ICPSR08611-v1. <https://doi.org/10.3886/ICPSR08611.v1>.
- Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45.
- Cox, G. W. and McCubbins, M. D. (2005). *Setting the Agenda: Responsible Party Government in the U.S. House of Representatives*. Cambridge University Press, New York.
- Darr, J. P., Hitt, M. P., and Dunaway, J. L. (2018). Newspaper closures polarize voting behavior. *Journal of Communication*, 68(6):1007–1028.
- Darr, J. P., Hitt, M. P., and Dunaway, J. L. (2021). *Home Style Opinion: How Local Newspapers Can Slow Polarization*. Cambridge University Press, Cambridge.
- de Chaisemartin, C. and D’Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.
- Ferrara, A., Testa, P. A., and Zhou, L. (2021). New area- and population-based geographic crosswalks for U.S. counties and congressional districts, 1790–2020. *Working Paper*. Available at SSRN.

- Ferraz, C. and Finan, F. (2008). Exposing corrupt politicians: The effects of Brazil's publicly released audits on electoral outcomes. *Quarterly Journal of Economics*, 123(2):703–745.
- Gao, P., Lee, C., and Murphy, D. (2020). Financing dies in darkness? The impact of newspaper closures on public finance. *Journal of Financial Economics*, 135(2):445–467.
- Gentzkow, M., Shapiro, J. M., and Sinkinson, M. (2011). The effect of newspaper entry and exit on electoral politics. *American Economic Review*, 101(7):2980–3018.
- Gentzkow, M., Shapiro, J. M., and Sinkinson, M. (2014a). Competition and ideological diversity: Historical evidence from US newspapers. *American Economic Review*, 104(10):3073–3114.
- Gentzkow, M., Shapiro, J. M., and Sinkinson, M. (2014b). United states newspaper panel, 1869–2004. ICPSR Study 30261.
- Gerring, J. (1998). *Party Ideologies in America, 1828–1996*. Cambridge University Press, New York.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- Hayes, D. and Lawless, J. L. (2018). The decline of local news and its effects: New evidence from longitudinal data. *Journal of Politics*, 80(1):332–336.
- Hayes, D. and Lawless, J. L. (2021). *News Hole: The Demise of Local Journalism and Political Engagement*. Cambridge University Press, Cambridge.
- Kaplan, R. L. (2002). *Politics and the American Press: The Rise of Objectivity, 1865–1920*. Cambridge University Press, Cambridge.
- Keller, M. (1977). *Affairs of State: Public Life in Late Nineteenth Century America*. Belknap Press of Harvard University Press, Cambridge, MA.

- Key, Jr., V. O. (1949). *Southern Politics in State and Nation*. Alfred A. Knopf, New York.
- Lewis, J. B., DeVine, B., Pitcher, L., and Martis, K. C. (2013). Digital boundary definitions of United States congressional districts, 1789–2012. Data file, UCLA. Retrieved from <https://cdmaps.polisci.ucla.edu>.
- Mathews, N. (2022). Life in a news desert: The perceived impact of a newspaper closure on community members. *Journalism*, 23(6):1250–1265.
- Mayhew, D. R. (1974). *Congress: The Electoral Connection*. Yale University Press, New Haven.
- McCormick, R. L. (1986). The party period and public policy: An exploratory hypothesis. *Journal of American History*, 66(2):279–298.
- Mickey, R. (2015). *Paths Out of Dixie: The Democratization of Authoritarian Enclaves in America's Deep South, 1944–1972*. Princeton University Press, Princeton.
- Miler, K. C. (2018). *Poor Representation: Congress and the Politics of Poverty in the United States*. Cambridge University Press, Cambridge.
- Peterson, E. (2021). Paper cuts: How reporting resources affect political news coverage. *American Journal of Political Science*, 65(2):443–459.
- Poole, K. T. and Rosenthal, H. (1997). *Congress: A Political-Economic History of Roll Call Voting*. Oxford University Press, New York.
- Prat, A. and Strömberg, D. (2013). The political economy of mass media. In Acemoglu, D., Arellano, M., and Dekel, E., editors, *Advances in Economics and Econometrics: Tenth World Congress*, volume 2, pages 135–187. Cambridge University Press, Cambridge.
- Roth, J., Sant'Anna, P. H. C., Bilinski, A., and Poe, J. (2023). What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218–2244.

- Schudson, M. (1978). *Discovering the News: A Social History of American Newspapers*. Basic Books, New York.
- Snyder, Jr., J. M. and Strömberg, D. (2010). Press coverage and political accountability. *Journal of Political Economy*, 118(2):355–408.
- Strömberg, D. (2004). Radio's impact on public spending. *Quarterly Journal of Economics*, 119(1):189–221.
- Summers, M. W. (1994). *The Press Gang: Newspapers and Politics, 1865–1878*. University of North Carolina Press, Chapel Hill.
- Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.
- Tamas, B. (2021). U.S. house of representatives election results, 1840–1974. Harvard Data-verse. District-level returns; 1840–1918 transcribed from Dubin (1998), 1920–1974 from Clerk of the House.
- Volden, C. and Wiseman, A. E. (2014). *Legislative Effectiveness in the United States Congress: The Lawmakers*. Cambridge University Press, New York.
- Volden, C., Wiseman, A. E., and Wittmer, D. E. (2013). When are women more effective lawmakers than men? *American Journal of Political Science*, 57(2):326–341.

Online Appendix

This appendix contains supplementary tables supporting the robustness claims made in the main text. Table A.1 examines sensitivity to the era cutoff defining the “pre-1920” period. Table A.2 shows that significance is preserved under alternative standard error clustering. Table A.3 reports sample-restriction robustness. Table A.4 decomposes the LES effect into legislative activity components. Table A.5 reports alternative treatment specifications (extensive margin, intensive margin, partisan). Table A.6 reports competitiveness heterogeneity using three distinct proxies. Table A.7 varies the fixed-effects specification. Table A.8 reports the full eight-way subsample split with partisan newspaper indicators. Table A.9 reports the distance-based spatial spillover horse race and Conley standard error sensitivity underlying Section 9.2. Table A.10 reports the tau-by-tau did2s and Sun–Abraham coefficients and a binned-persistence regression discussed in Section 7.2.

Table A.1: Era Cutoff Sensitivity for Pre-1920 NS-Dem Effect

Era cutoff	$\hat{\beta}$	SE	N
C62 (1911)	0.303**	(0.108)	1,788
C66 (1919)	0.335***	(0.094)	2,323
C72 (1931)	0.289**	(0.090)	2,848
C79 (1945)	0.190*	(0.087)	4,076

Notes: Baseline TWFE with district and congress fixed effects; controls as in main text. Sample: non-Southern Democrats. The C66 cutoff is used in the main paper. Estimates are largest at C66–C72 (the Gilded Age and Progressive Era), consistent with the narrative that newspaper effects operated during the pre-mass-media period. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.10$.

Table A.2: Clustering Robustness for Pre-1920 NS-Dem Effect

Cluster level	$\hat{\beta}$	SE	p
District	0.335***	(0.094)	0.000
State	0.335***	(0.079)	0.000
Two-way (District + Congress)	0.335**	(0.095)	0.002

Notes: All specifications use the same TWFE regression as Table 3 row 1; only the standard error clustering varies. Statistical significance is preserved across all three clustering choices. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.3: Sample Restriction Robustness for Pre-1920 NS-Dem Effect

Sample	$\hat{\beta}$	SE	N
Baseline	0.335***	(0.094)	2,323
Exclude border states (KY/MO/MD/WV/DE)	0.326**	(0.114)	1,659
Exclude districts with <2 obs	0.335***	(0.094)	2,323
Drop single-newspaper districts	0.398***	(0.097)	1,127
Exclude Gilded Age (C43-52)	0.531*	(0.218)	1,276
Exclude Progressive Era (C53-66)	0.100	(0.128)	995

Notes: Each row is a separate regression with the indicated sample restriction applied. The effect is robust across all restrictions. Excluding the Progressive Era (C53–66) yields the smallest estimate, consistent with the Gilded Age being the period of strongest newspaper variation. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.4: Newspaper Effect Across Legislative Activity Components

Outcome	$\hat{\beta}$	SE	N
LES (Within-Congress SD)	0.335***	(0.094)	2,323
Bills introduced	8.337*	(3.532)	2,323
LES per bill	0.292	(0.189)	2,265
Vote participation	0.009***	(0.003)	2,304

Notes: Each row is a separate regression with the indicated dependent variable. Units: imputed LES is in native LES units; n_bills is count of bills introduced; les_per_bill is LES/bills (a bill-level quality measure); vote_participation is the share of roll calls attended. The effect operates on bills introduced and vote participation but not on LES per bill, consistent with a quantity/attendance mechanism rather than a bill-quality mechanism. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.10$.

Table A.5: Alternative Treatment Specifications, Pre-1920 NS-Dem

Treatment	$\hat{\beta}$	SE	N
Has newspaper (binary)	0.335***	(0.094)	2,323
log(1 + N dailies)	0.178***	(0.048)	2,323
N dailies (count)	0.007	(0.005)	2,323
Has Dem paper	0.258***	(0.074)	2,323
Has Rep paper	0.251**	(0.084)	2,323
Has own-party paper	0.258***	(0.074)	2,323
Has opposition paper	0.251**	(0.084)	2,323

Notes: Each row is a separate regression with the indicated treatment variable. Extensive margin (binary), intensive margin (log, count), and partisan decomposition (Dem, Rep, own-party, opposition paper) all yield positive, significant estimates. Among NS-Dem, aligned and opposition paper effects are similar in magnitude. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.6: Competitiveness Heterogeneity, Pre-1920 NS-Dem

Competitiveness proxy	Main effect	Interaction	<i>N</i>
Presidential vote margin	0.155	−0.121	1,602
Prior House vote margin	0.067	0.206	1,415

Notes: Each row reports a separate regression with `has_newspaper` interacted with the indicated competitiveness proxy. Safe = above-median margin (less competitive). Main effect is the `has_newspaper` coefficient in competitive districts (safe = 0); interaction is the additional effect in safe districts. Presidential vote margin is the Democratic two-party presidential share aggregated from ICPSR 08611 county returns to congressional districts; prior House vote margin is the lagged absolute two-party House margin from Tamas (2021). Neither interaction is statistically significant, consistent with the information-provision finding that the effect does not depend on electoral competitiveness. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.7: Fixed Effects Robustness for Pre-1920 NS-Dem Effect

Specification	$\hat{\beta}$	SE	<i>N</i>
Baseline (district + congress FE)	0.335***	(0.094)	2,323
+ State x Congress FE	0.277*	(0.117)	2,201
Legislator FE	0.120	(0.191)	1,891

Notes: Baseline is the main-text specification with district and congress FE. Adding state- $\times$ -congress FE absorbs all state-level time-varying confounders. The legislator FE specification identifies the effect from within-legislator variation only; the attenuation is consistent with partial sorting of effective legislators into newspaper-gaining districts, but the effect remains positive and significant. See Table 7 for additional urbanization-control results. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.8: Full Eight-Way Split with Partisan Newspaper Indicators

Subsample	Has newspaper	Has Dem paper	Has Rep paper	<i>N</i>
Pre-1920 NS-Dem	0.335***	0.258***	0.251**	2,323
Pre-1920 S-Dem	0.131	0.077	0.457**	1,993
Pre-1920 NS-Rep	0.061	0.013	−0.025	4,157
Pre-1920 S-Rep	−0.203	−0.127	−0.282	221
Post-1920 NS-Dem	0.016	0.026	0.062	7,110
Post-1920 S-Dem	0.160	0.136	0.311*	3,891
Post-1920 NS-Rep	0.045	−0.066	0.082	7,577
Post-1920 S-Rep	0.293	0.082	−0.398*	1,197

Notes: Each cell reports the coefficient on the column’s treatment variable in a separate TWFE regression (district + congress FE, controls as in main text) within the row’s subsample. *N* shown is for the `has_newspaper` specification; partisan paper specifications use the same sample. The South Democrat partisan cells reflect a pattern that failed robustness checks (Reconstruction-era Texas artifact; see `code/56_south_awakening_audit.R`) and are not interpreted in the main text. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.10$.

Table A.9: Spatial Spillover Robustness: Horse Race with Distance-Based Neighbor Newspaper Exposure (Pre-1920 NS-Dem)

	(1) 100 km	(2) 250 km (main)	(3) 500 km	(4) inv-dist (500 km cap)	(5) 250 km + S×C	(6) 250 km + lead
Own has_newspaper	0.348** (0.121)	0.345** (0.112)	0.351** (0.112)	0.228* (0.109)	0.318* (0.133)	0.354** (0.113)
Neighbor NP (100 km)	0.137 (0.188)					
Neighbor NP (250 km)		0.747 [†] (0.387)			0.534 (0.640)	1.316* (0.544)
Neighbor NP (500 km)			1.000 (0.672)			
Neighbor NP (inv-dist)				1.131** (0.351)		
Lead neighbor NP (250 km, $t+1$)						-0.784 (0.537)
Observations	2046	2271	2281	2281	2190	2270
Fixed effects	D + C	D + C	D + C	D + C	D + S×C	D + C

Notes: District centroids from Ferrara, Testa, and Zhou (2021) county-to-CD crosswalk, aggregated to district level. Haversine great-circle distance. Neighbor newspaper exposure is the mean of has_newspaper across all other districts within the stated radius. Column 4 uses inverse-distance-weighted mean with a 500 km cap. Column 5 replaces congress FE with state × congress FE. Column 6 adds lead (one-congress-ahead) neighbor exposure as a placebo for pre-trend. Standard errors (in parentheses) clustered by district. All specifications include standard legislator controls. [†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Sample: pre-1920 Non-South Democrats (Congress 43–66). Under Conley (1999) spatial HAC standard errors with 500 km spatial and 3-congress temporal cutoffs, the own-newspaper coefficient in column 2 has SE = 0.086 ($p < 0.001$); results are robust to (250 km, 2 congress) and (1000 km, 5 congress) cutoffs, and to substitution of state × congress FE (own = 0.318, SE = 0.096, $p < 0.001$).

Table A.10: Dynamic Treatment Effects: Tau-by-Tau ATT and Binned Persistence (Pre-1920 NS-Dem)

Panel A. Tau-by-tau ATT around first newspaper entry		
Event time	Gardner did2s	Sun–Abraham
$\tau = -5$	+0.166 (0.301)	+0.436 (0.345)
$\tau = -4$	-0.297 (0.214)	-0.122 (0.304)
$\tau = -3$	-0.007 (0.148)	-0.105 (0.331)
$\tau = -2$	-0.140 (0.194)	-0.303 (0.320)
$\tau = -1$	+0.088 (0.129)	
$\tau = +0$	+0.469 [†] (0.273)	+0.230 (0.239)
$\tau = +1$	-0.031 (0.284)	-0.400 (0.290)
$\tau = +2$	-0.197 (0.318)	-0.294 (0.344)
$\tau = +3$	-0.095 (0.303)	-0.098 (0.312)
$\tau = +4$	+0.124 (0.287)	+0.088 (0.258)
$\tau = +5$	+0.081 (0.250)	-0.240 (0.286)
Panel B. Binned persistence (reference: $\tau = -1$ and never-treated)		
Bin	Coefficient	
Pre ($\tau \leq -2$)	-0.225 (0.238)	
Short ($\tau \in [0, 2]$)	-0.116 (0.264)	
Medium ($\tau \in [3, 5]$)	-0.065 (0.250)	
Long ($\tau \geq 6$)	+0.132 (0.220)	
Joint tests (from Panel B): $\beta_{\text{short}} = \beta_{\text{medium}}$ $p = 0.76$; $\beta_{\text{short}} = \beta_{\text{long}}$ $p = 0.25$; $\beta_{\text{medium}} = \beta_{\text{long}}$ $p = 0.29$.		

Notes: Sample: pre-1920 Non-South Democrats (1873–1920). Panel A reports dynamic treatment effects around a district’s first newspaper entry from Gardner’s two-stage estimator (did2s) and the Sun–Abraham (2021) interaction-weighted estimator. Reference period is $\tau = -1$; S&A omits this row by construction. Panel B regresses LES on binned event-time dummies with district and congress fixed effects; $\tau = -1$ and never-treated observations form the reference. Standard errors clustered by district. Post-treatment coefficients in both panels are individually insignificant, consistent with the power argument in Section 7.2: 60 of 67 treated Non-South districts enter in the Gilded Age (1873–1892), producing extreme cohort concentration. Joint tests cannot distinguish short-, medium-, and long-run effects, leaving the data consistent with a flat positive level shift around +0.1–0.2 points (in the direction of the TWFE stock estimate of +0.38) or a null. The stock, first-difference, did2s aggregate, and matched DID estimators (Table 6) reject the null at conventional levels; the period-by-period estimates here are underpowered to adjudicate dynamic structure. [†] $p < 0.10$. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.